

A comparison of the major open source compilers

Compiler/ SDK	Backer	Language	Distinguishing focus
Qiskit (Transpiler)	IBM	Python	Broadest feature set: pulse control, dynamic circuits, noise-aware passes, largest hardware install base
Cirq	Google	Python	Fine-grained control for NISQ research; native path to Google superconducting hardware
tket/pytket	Quantinuum (CQC)	C++/Python	Retargetable optimiser prized for aggressive circuit-depth and gate-count reduction
PennyLane + Catalyst	Xanadu	Python	JIT compilation for differentiable, hybrid quantum-classical programs; strong QML fit
XACC	Oak Ridge Nat'l Lab	C++	Extensible, HPC-oriented compiler for integrating quantum kernels into classical supercomputing workflows
Staq/ ProjectQ	Academic/ Community	C++/Python	Lightweight, research-friendly toolchains for circuit synthesis and translation between formats

sequences to lower the circuit depth and the overall number of gates using algebraic identities.

- Qubit mappings and routing — Logical qubits are linked to physical qubits, and SWAP gates are placed to ensure that two-qubit gates only connect physically neighbouring qubits.
- Basis gate decomposition — Decomposing arbitrary gates in terms of the gate set that a particular machine supports.
- Noise-aware scheduling — This is about reducing error by using calibration data, which includes error rates, gate durations and crosstalk, to identify the least noisy qubits and the optimal gate order. The output is usually a hardware-executable circuit, which is often accompanied by detailed, pulse-level, instructions for the control electronics.

Cirq (Google): Cirq is built for experimentation on near-term hardware and gives users flexibility in control, which is Google's superconducting processor. Until recently, access to Google's hardware was more limited than IBM's, which has kept its mindshare somewhat behind Qiskit.

tket/pytket (Quantinuum): Unlike Cirq and Qiskit, tket is a modular compiler that includes optimisation passes, which produce very thin circuits with low gate counts. It is easy to incorporate tket into other libraries.

PennyLane and Catalyst (Xanadu): Instead of focusing on optimisation, PennyLane has built tools for quantum machine learning. Its Catalyst just-in-time compiler now compiles directly to hardware control signals and is one of the first fully

open source stacks spanning software through hardware.

XACC, StaQ, ProjectQ, and more: Past the four main SDKs, a lengthier tail of community and lab-driven efforts targets unique use cases. XACC aims to facilitate integration of quantum kernels within classical high performance-computing frameworks, while StaQ and ProjectQ provide circuit synthesis and format translation tools that are more appropriate for use in academia.

Core compilation techniques

Gate-level optimisation: Here, various peephole optimisation rewrite rules are applied to circuits. Gates are cancelled by their inverse and moved between circuits, while rotations are merged. To the best of my knowledge, no hardware mapping is employed at this level.

The major open source compilers

A handful of projects dominate current usage, each shaped by the hardware vendor or lab behind it.

Qiskit Transpiler (IBM): Qiskit is the most feature-rich open source library, covering the entirety of circuit construction, control, and dynamic circuits as well as modelling, noise, and simulation. It also has tight integration with IBM hardware.

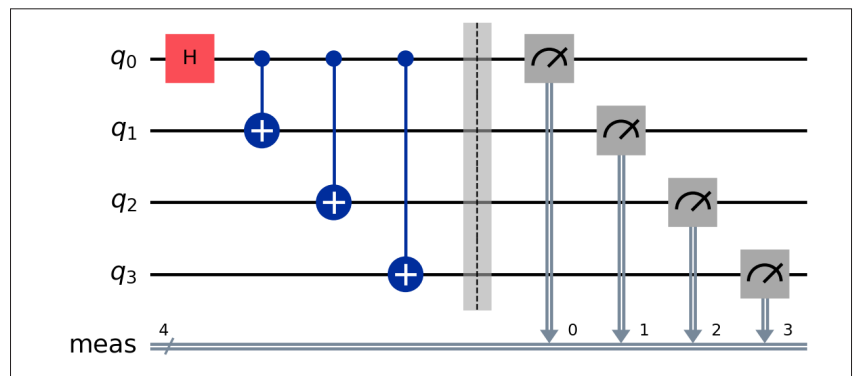


Figure 1: The logical circuit as written, before compilation